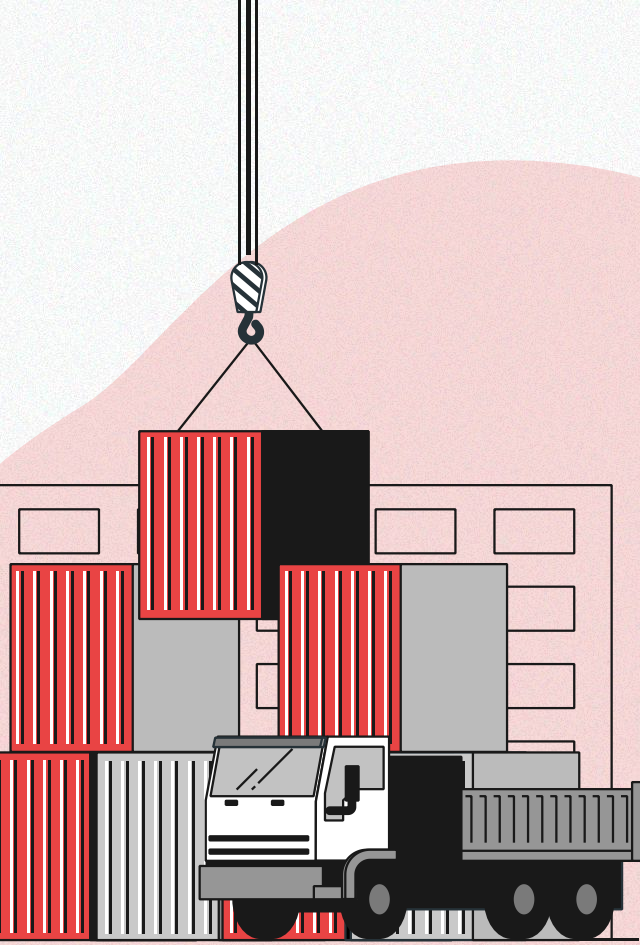




CAPSTONE PROJECT: ONLINE RETAIL CUSTOMER ANALYTICS & SEGMENTATION

Data-Driven Customer
Insights & K-Means Clustering

Prepared by: Abdallah

PROJECT OBJECTIVE & DATASET OVERVIEW

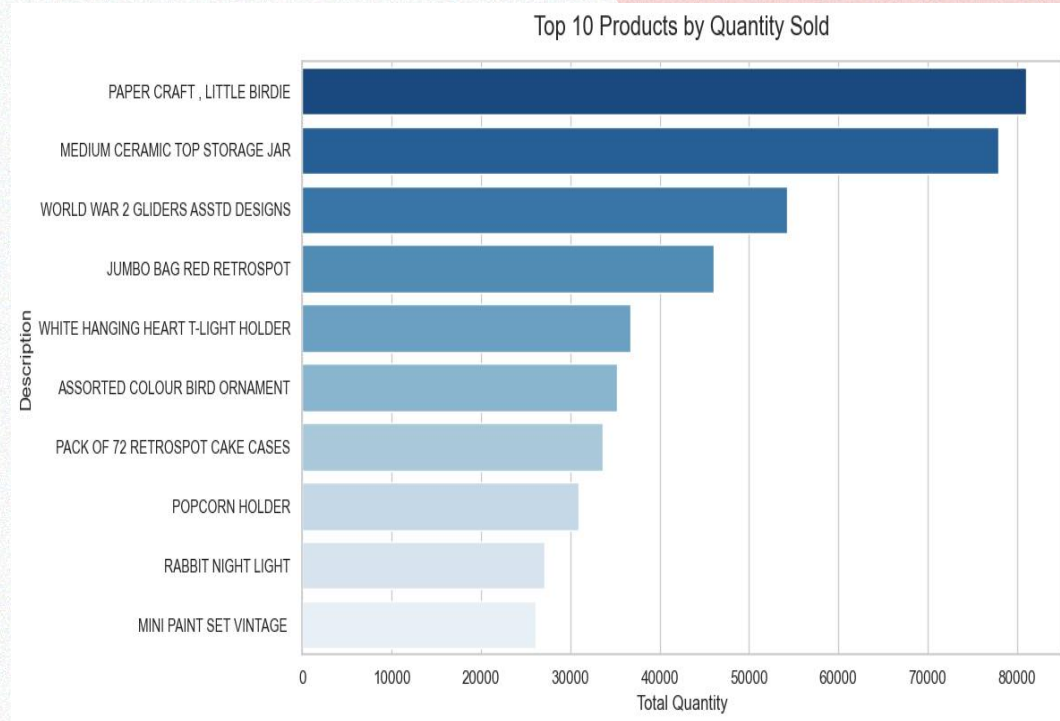
- Objective:** Act as Data Analysts to perform deep Exploratory Data Analysis (EDA), extract actionable business insights, and build data-driven customer segments.
- Dataset:** UCI Online Retail Dataset (Dec 2010 - Dec 2011).
- Data Cleaning Decisions:**
 1. Converted InvoiceDate to datetime.
 2. Dropped transactions missing CustomerID (critical for segmentation).
 3. Removed exact duplicates and filtered out cancellations (negative quantities/prices) to ensure pure sales analysis.

**EXPLORATORY DATA
ANALYSIS - PRODUCT &
MARKET INSIGHTS**

TOP PRODUCTS:

* *BY QUANTITY:*

-“PAPER CRAFT ,LITTLE BIRDIE”
-(HIGH-VOLUME, LOW-MARGIN
DRIVER).

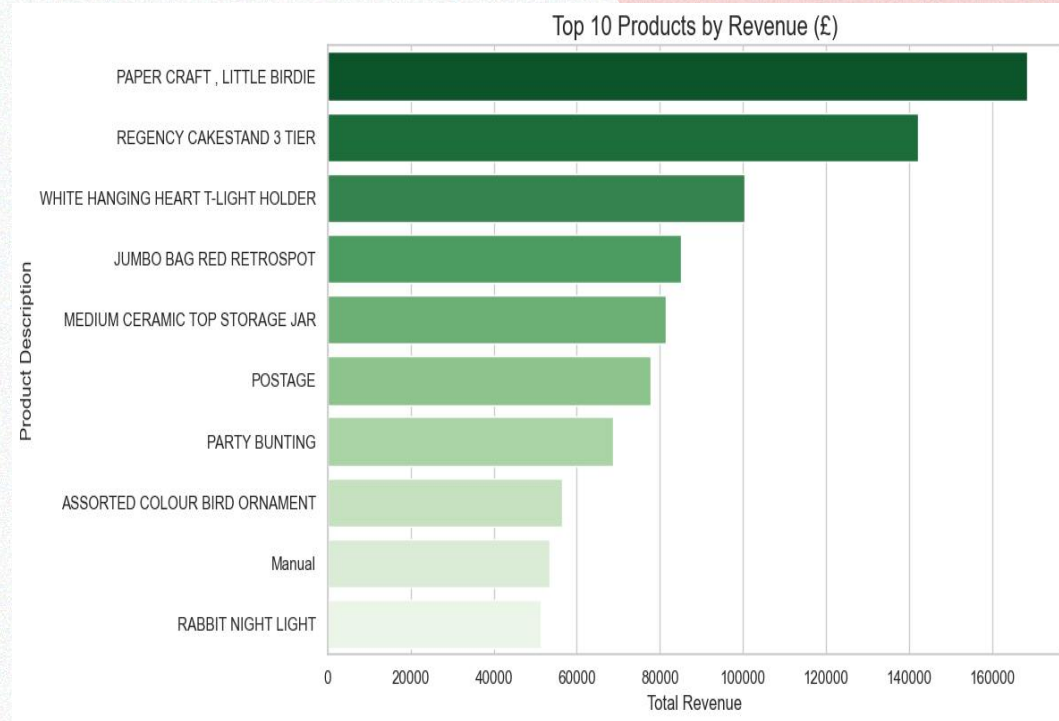


TOP PRODUCTS:

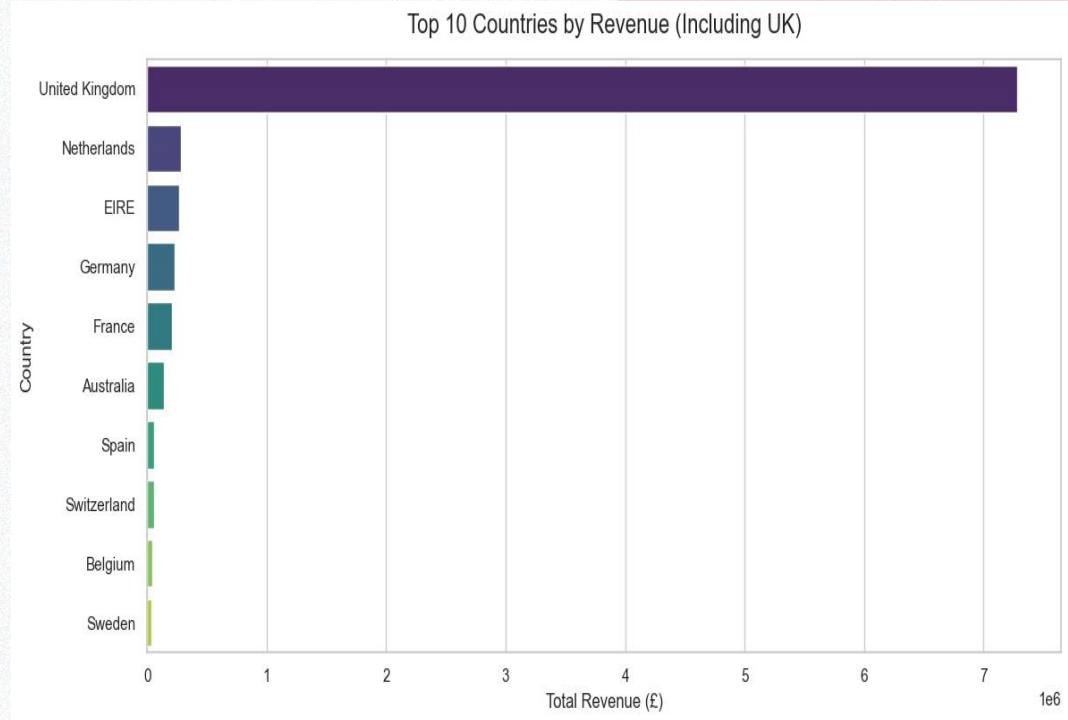
** BY REVENUE:*

- " PAPER CRAFT ,LITTLE BIRDIE "

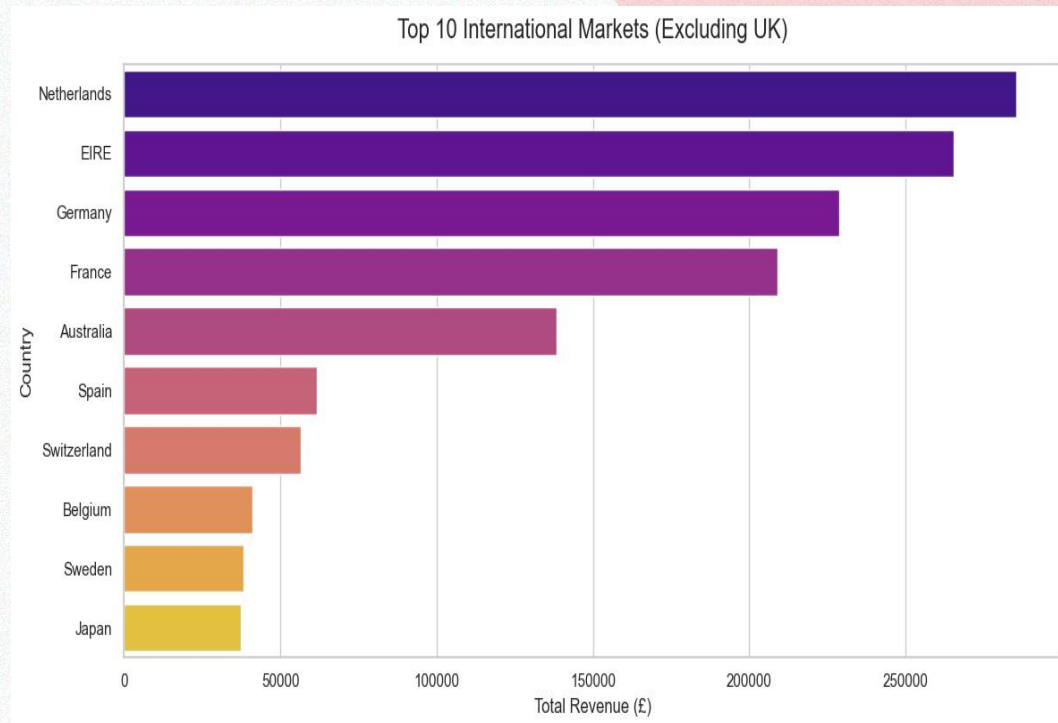
-(THE HERO PRODUCT GENERATING
MAXIMUM REVENUE).



**GEOGRAPHICAL
DOMINANCE:
THE UK DOMINATES OVER
90% OF TOTAL REVENUE.**



❏ GEOGRAPHICAL DOMINANCE:
INTERNATIONAL MARKETS
(EXCLUDING UK):
NETHERLANDS ,EIRE AND GERMANY
ARE THE TOP 3 REVENUE-
GENERATING FOREIGN MARKETS.



**EXPLORATORY DATA
ANALYSIS - TEMPORAL
SALES TRENDS**



SEASONALITY (Q4 PEAK)

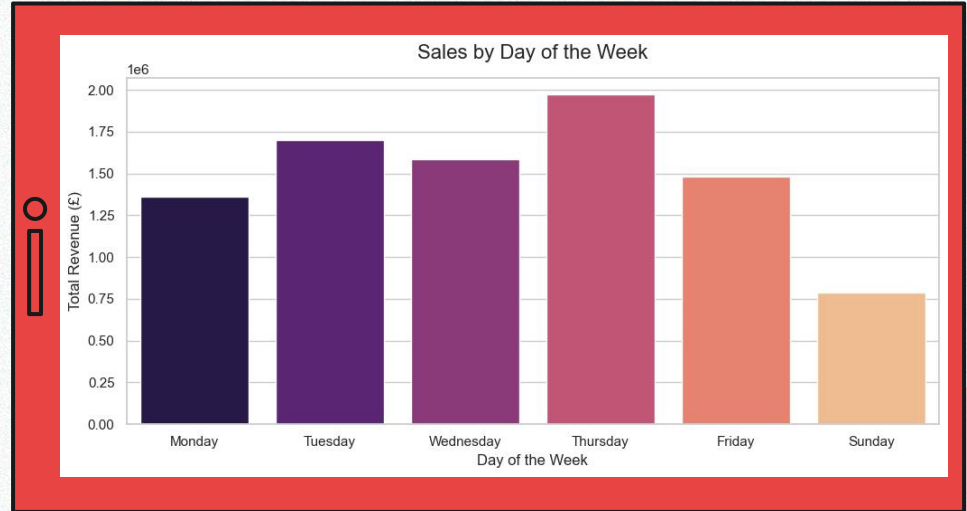
- ❑ A massive revenue surge in September, October, November, and December due to Q4 holiday and Christmas shopping

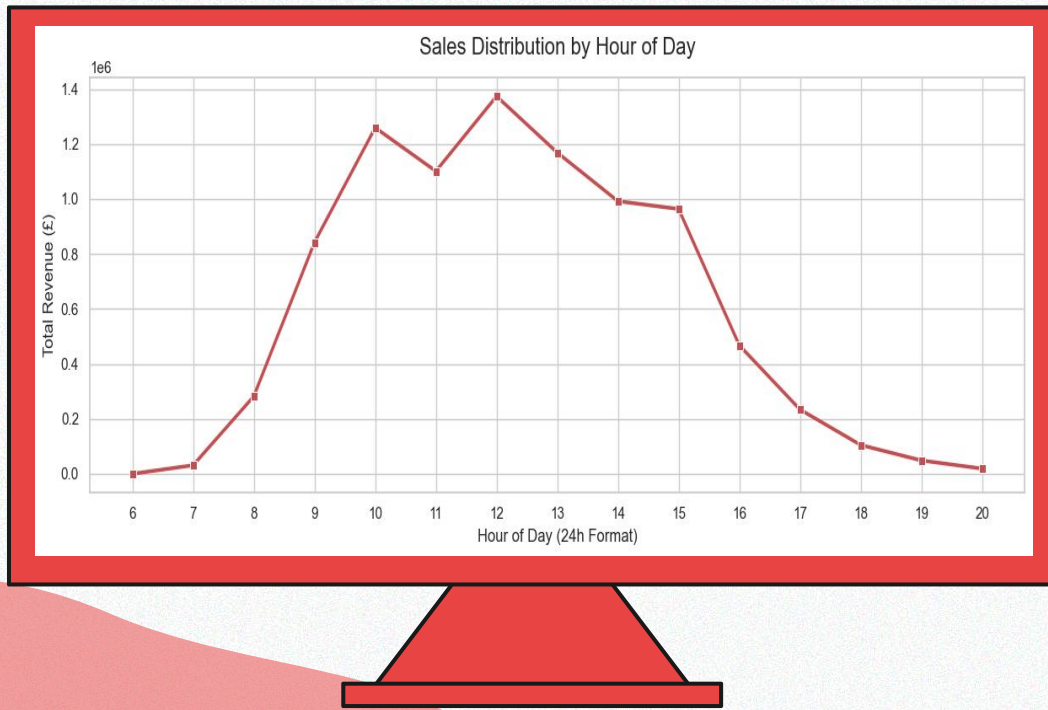
SEASONALITY (Q4 PEAK)



OPERATIONAL BLINDSPOT (THE SATURDAY MYSTERY):

- ❑ Zero transactions recorded on Saturdays, indicating a potential revenue loss due to weekend closure.



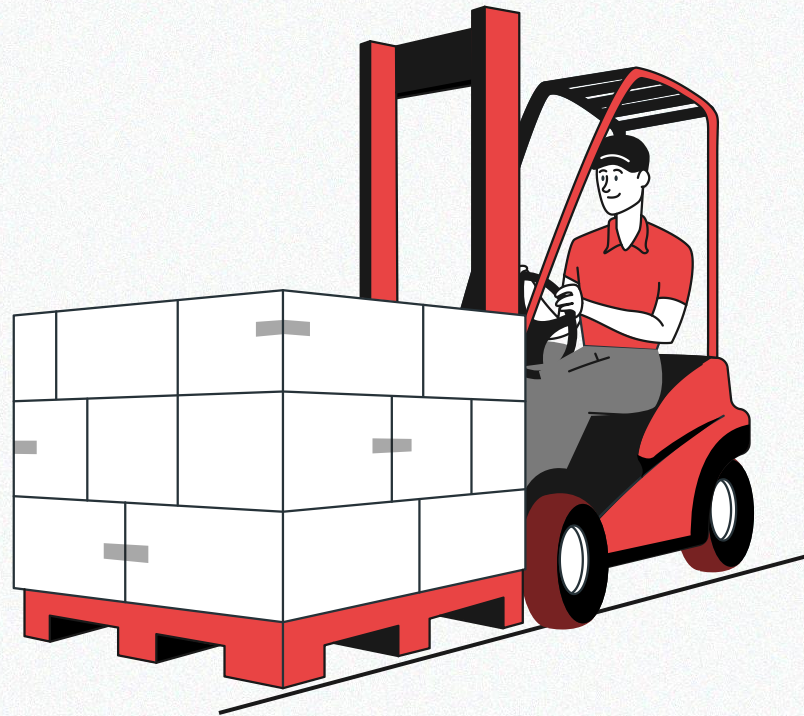


PEAK HOURS:

- ❑ Sales hit global maximum between **10:00 AM** and **12:00 PM**, defining the golden window for marketing outreach

(RFM)

CUSTOMER SEGMENTATION METHODOLOGY



FEATURE ENGINEERING:

FRAMEWORK BASED ON RFM ANALYSIS



Recency (R)

- Days since the last purchase.



Frequency (F)

- Total number of unique orders.



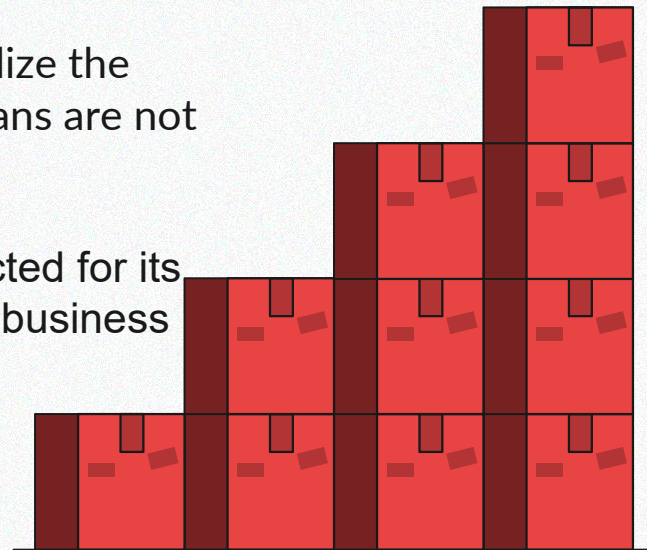
Monetary (M)

Total spending per customer.

FEATURE ENGINEERING:

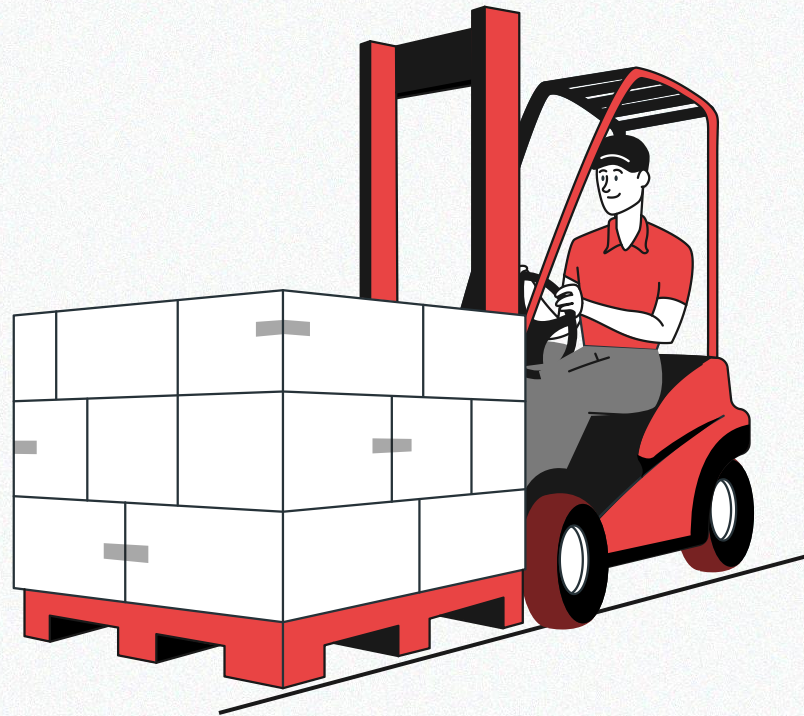
FRAMEWORK BASED ON RFM ANALYSIS

- Preprocessing:** Applied **StandardScaler** to normalize the RFM metrics, ensuring distance calculations in K-Means are not skewed by different scales.
- Algorithm Choice: K-Means Clustering** was selected for its high scalability on large transactional data and clear business interpretability.

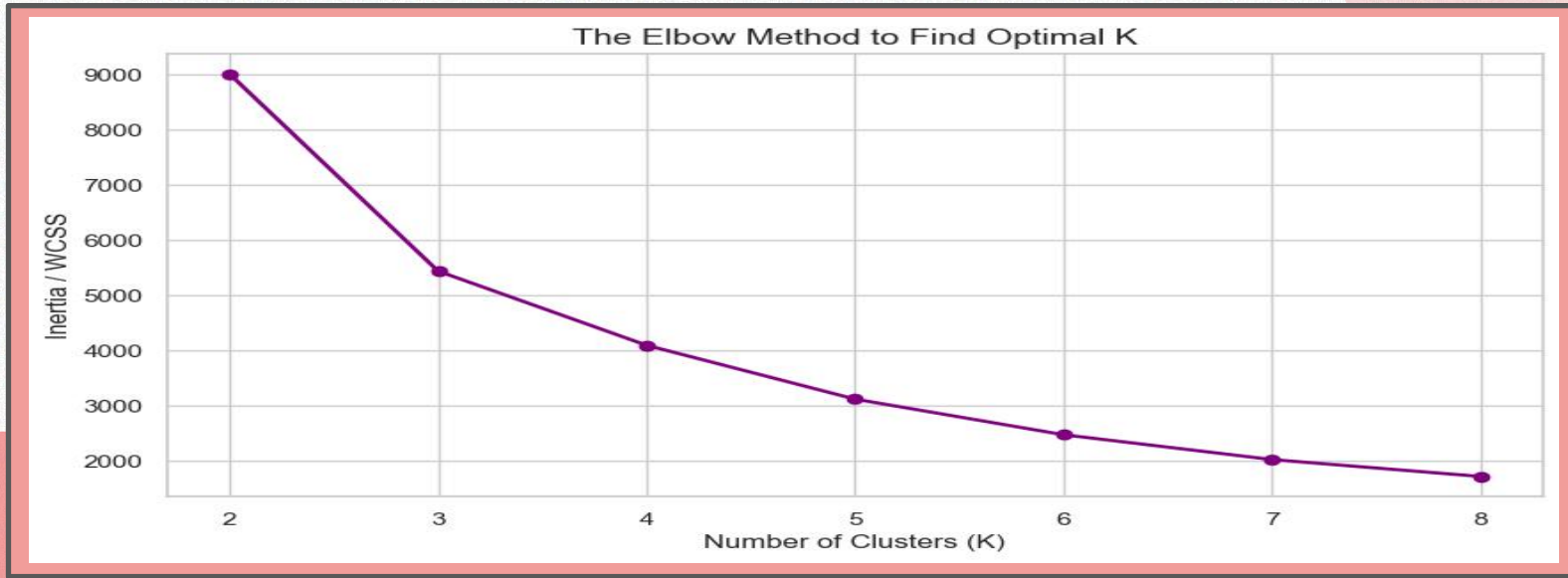


THE ELBOW METHOD

MACHINE LEARNING
OPTIMIZATION



THE ELBOW METHOD



- **Optimal K Selection:** Evaluated WCSS (Inertia) from $K=2$ to $K=8$.
- **The Elbow Point:** A clear inflection point is observed at $K = 4$, providing the most stable and balanced variance reduction.
- **Stability:** Used $n_{init}=10$ to guarantee centroid convergence across multiple random initializations.

THE 4 CUSTOMER SEGMENTS (THE CORE OUTPUT)

01. VIP / CHAMPIONS

Recency: ~7 days | Frequency: ~82 | Monetary:
~£127K. (The highest revenue drivers).

02. LOYAL CUSTOMERS

Recency: ~15 days | Frequency: ~22 |
Monetary: ~£12.6K. (Highly frequent, high-trust buyers).

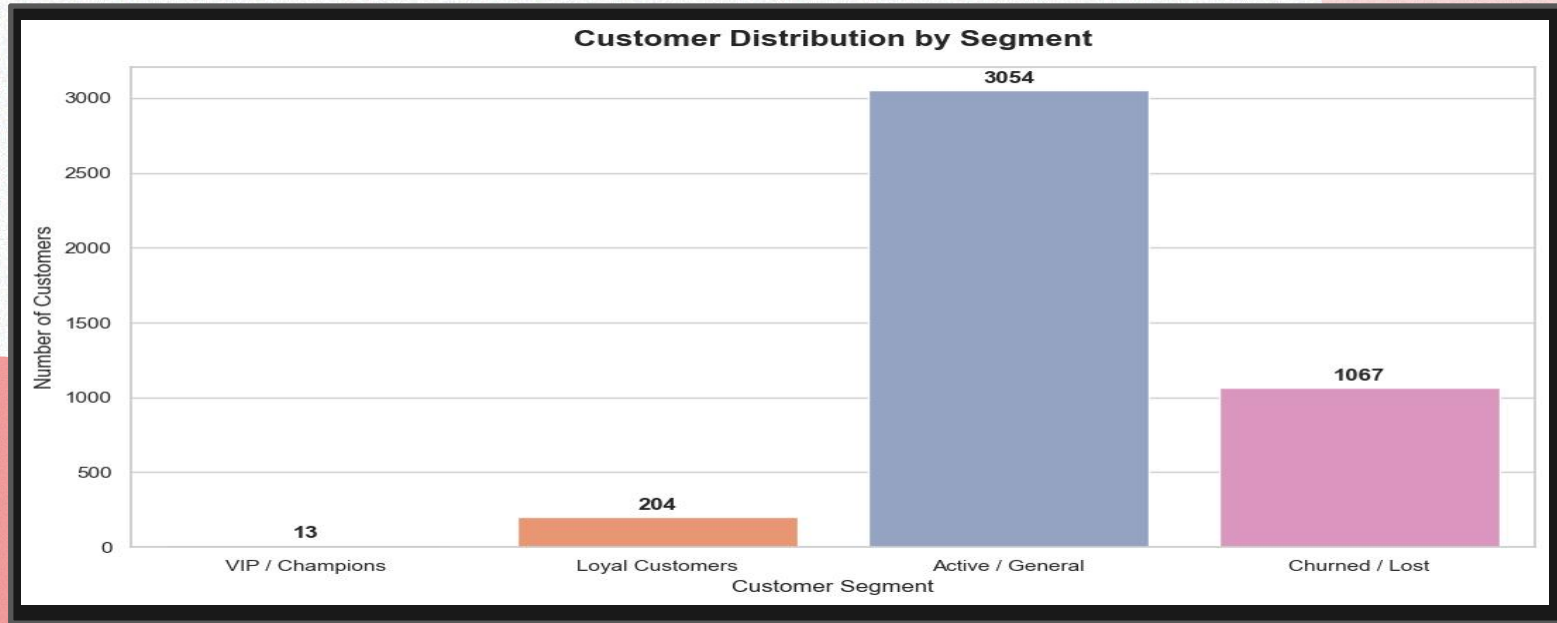
03. ACTIVE / GENERAL MARKET

Recency: ~43 days | Frequency: ~3.6 | Monetary:
~£1.3K. (The core volume/mass market).

04. CHURNED / LOST CUSTOMERS

Recency: ~248 days | Frequency: ~1.5 | Monetary:
~£478. (Inactive for >8 months).

THE 4 CUSTOMER SEGMENTS (THE CORE OUTPUT)





BUSINESS RECOMMENDATIONS & CONCLUSION

VIPS

Implement dedicated account managers and exclusive rewards to prevent churn

GENERAL MARKET

Push seasonal discounts and targeted newsletters during peak hours (10 AM - 3 PM) to increase transaction frequency

LOYAL BUYERS

Earth's isLaunch cross-selling campaigns for new inventory arrivals. the Sun's mass

SUPPLY CHAIN

Secure peak inventory for Hero products before the massive Q4 seasonal surge.

THANKS!

Do you have any questions?

Abdullah

